

Customer Churn Analysis

Telecom Subscription Data — SQL, Excel & Power BI Project

Data cleaning · SQL analysis · Power BI dashboard · Business insights

1. Project Overview

This project analyzes customer churn for a telecom subscription business. Raw customer data was cleaned and structured, explored using SQL, and visualized in an interactive Power BI dashboard to identify who is churning, why, and where the business should focus retention efforts.

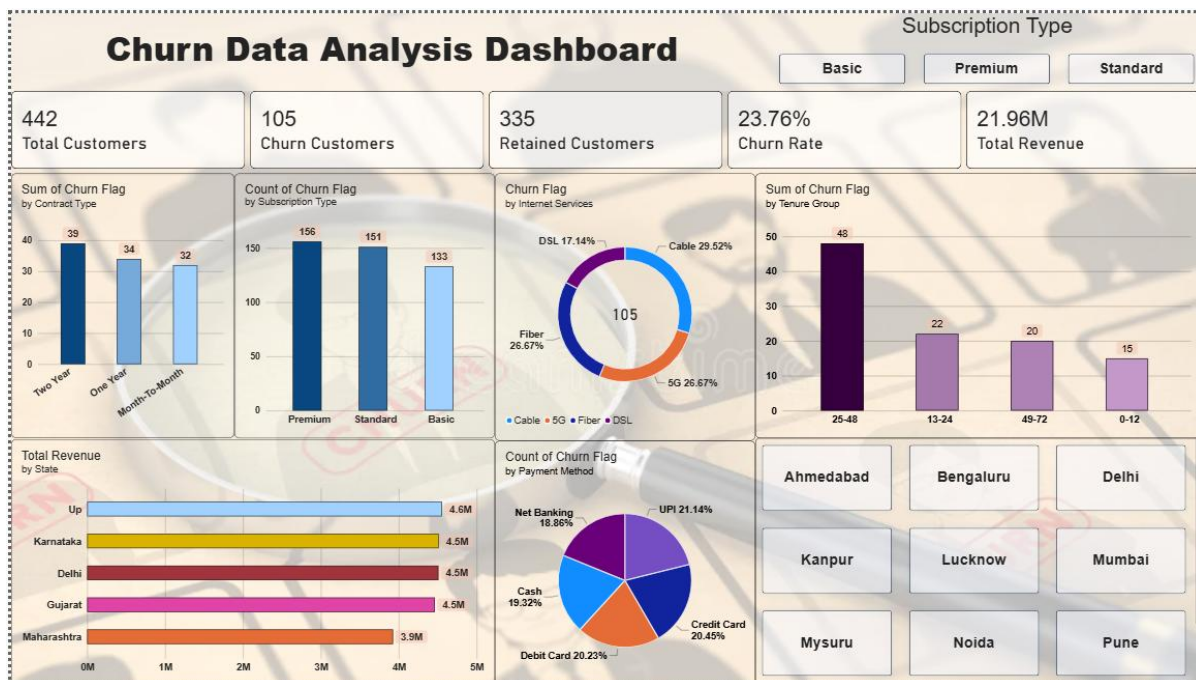
Tools used: Excel (data cleaning) · SQL (querying & analysis) · Power BI (dashboard) · Jupyter Notebook (exploration)

2. Key Metrics

442 Total Customers	105 Churned Customers	335 Retained Customers	23.76% Churn Rate	₹21.96M Total Revenue
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3. Power BI Dashboard

The dashboard below breaks down churn by contract type, subscription plan, internet service, tenure, state, and payment method.



4. Key Insights

Contract & Subscription

- Two-Year contracts churn the most (27.97%), more than Month-to-Month (20.6%) — long-term commitment isn't preventing churn.
- Standard and Premium subscribers churn roughly 2x more than Basic subscribers.
- High-value customers are at higher churn risk.

Service Quality

- Cable internet has the highest churn rate (31.19%).
- DSL has the lowest churn rate (16.51%).
- Service quality appears to meaningfully impact retention.

Customer Lifecycle

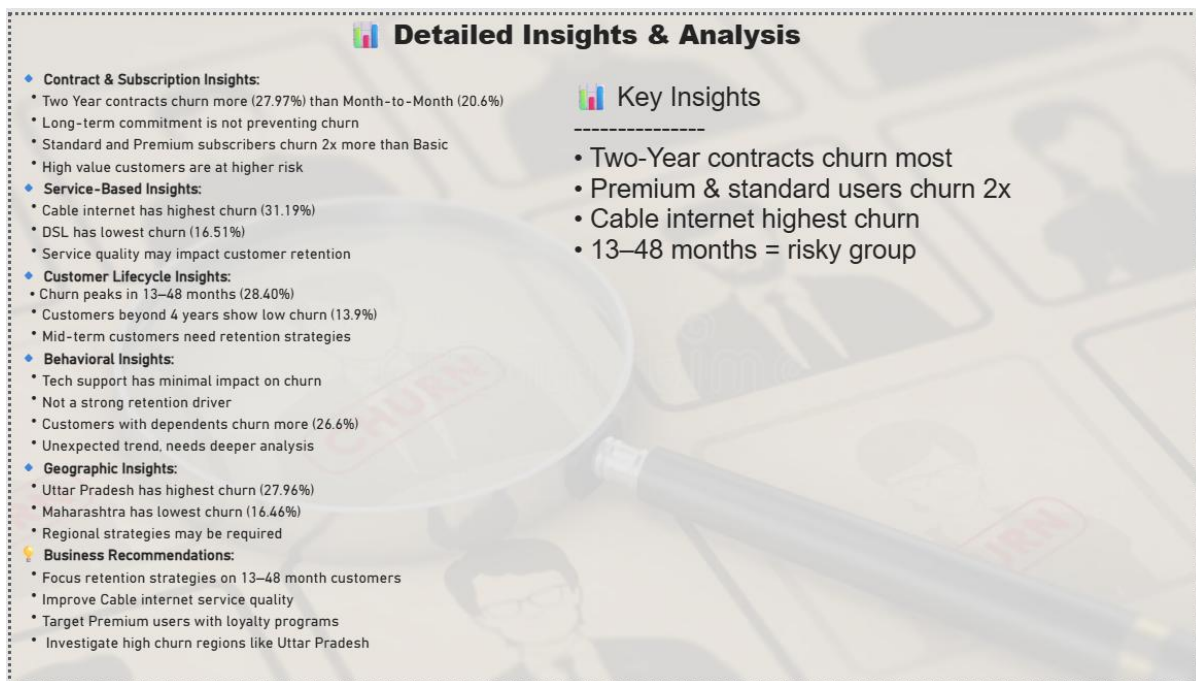
- Churn peaks in the 13–48 month tenure range (28.40% of churned customers).
- Customers beyond 4 years show low churn (13.9%).
- Mid-term customers are the priority group for retention strategies.

Behavioral Patterns

- Tech support usage has minimal impact on churn — not a strong retention driver.
- Customers with dependents churn more (26.6%), an unexpected trend worth deeper analysis.

Geographic

- Uttar Pradesh has the highest churn rate (27.96%).
- Maharashtra has the lowest churn rate (16.46%).
- Regional retention strategies may be required.



5. Business Recommendations

- Focus retention campaigns on customers in the 13–48 month tenure window.

- Investigate and improve Cable internet service quality.
- Target Premium and Standard subscribers with loyalty programs.
- Investigate high-churn regions such as Uttar Pradesh for local causes.
- Re-examine why customers with dependents churn more, and design targeted offers for this segment.

6. SQL Analysis Approach

The cleaned dataset was loaded into a SQL database (customer_churn table) and queried to calculate churn rate, averages, and breakdowns by segment. Sample queries used:

Analysis	Purpose
Overall churn rate	% of customers who churned, out of total
Churn by contract type	Compare churn across Month-to-Month / One Year / Two Year
Churn by internet service	Compare churn across DSL / Fiber / Cable / 5G
Churn by state	Identify highest-churn regions
Average monthly charges & tenure	Baseline customer profile metrics
Top 10 high-value customers	Ranked by customer_value
Customers without tech support	Segment for churn-driver review

7. Repository Contents

File	Description
Churn_Unclean_Project.xlsx	Raw, unprocessed source data
Clean_Churn_Data.csv	Cleaned dataset used for analysis (442 customers)
Customer_churn_analysis.sql	SQL queries used to analyze churn drivers
Churn_Analysis.ipynb	Jupyter notebook for exploratory data analysis
Churn_Data_Analysis.pbix	Power BI dashboard file
Churn_analysis_dashboard.png	Dashboard screenshot
Churn_Analysis_insights.png	Summary of key insights

8. Conclusion

Churn is concentrated among mid-tenure, higher-tier, Cable-internet customers, and varies notably by region. Retention efforts targeted at the 13–48 month window, Premium/Standard subscribers, and high-churn states like Uttar Pradesh offer the clearest opportunity to reduce the current 23.76% churn rate.